

UTKARSH SINGH

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Professional Summary

Second-year Computer Science Engineering student specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning.

Education

Maharaja Agrasen Institute of Technology [GGSIPU] New Delhi, India
B.Tech in Computer Science Engineering, Specialization in Artificial Intelligence Aug. 2025 – May 2029
(9.3 CGPA)

Kamal Public Sr. Sec. School [CBSE] New Delhi, India
Class XII PCM with Computer Science (92.4%) April 2011 – May 2025

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C++, C

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, PyTorch, TensorFlow, Transformers, Hugging Face, scikit-learn

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs **Frontend:** React, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, SQLite

MLOps & Cloud: Docker, Git, GitHub Actions (CI/CD), AWS (S3, EC2, Lambda), GCP, Vercel, Render, Railway

Libraries & Tools: NumPy, Pandas, LangChain, Matplotlib, OpenCV, Postman, Jupyter

Projects

KAVACH — Fraud Risk Manager for Indian D2C Merchants

- Python, FastAPI, XGBoost, SHAP, NetworkX, Next.js — built a calibrated return-risk scorer paired with graph-based fraud ring detection over shared phones, devices, VPAs and fuzzy-matched addresses.
- Defence-only risk manager for Indian D2C merchants who lose 5–15% of gross margin to COD return abuse and coordinated chargebacks; ring detection scored 0.889 precision at 1.000 recall on held-out data.

NETRA — AI Growth Copilot for Indian Kirana Stores

- Python, FastAPI, Llama 3.3 70B, Sarvam AI, React, Differential Privacy — built demand forecasting, non-collusive price benchmarking and credit scoring delivered via Soundbox audio, WhatsApp and web in 7 Indic languages.
- AI growth copilot for Indian Kirana stores serving 90% of a \$600B+ grocery market; enforces a five-layer privacy boundary using k-anonymity cohort gating and Laplace DP to block competitor surveillance.

ATLASS — Adaptive Taste Learning And Suggestion System

- JavaScript, SVD, LSA/TF-IDF, WebGL, GSAP, TMDb API — engineered SVD fold-in blended 70/30 with a 48-dimensional LSA content profile, pre-trained weights shipped as ~700KB JSON, re-scoring in under 50ms.
- Hybrid movie recommender running entirely in-browser with zero inference server; built on MovieLens (9,742 films, 100K+ ratings) with a WebGL gallery rendering cards along a parametric arc.

SCOUT — AI-Powered Cross-Platform Shopping Assistant

- Python, FastAPI, Claude API, SSE, Redis, SerpAPI — built an LLM-based intent router handling Hindi/Hinglish with slang and typos, per-session Redis state, and token-by-token SSE streaming.
- AI shopping agent that concurrently searches seven Indian e-commerce platforms, deduplicates results by title fingerprint, and ranks products on a transparent six-factor weighted score.

Achievements

Top 10 Finalist — Paytm Build for India AI Hackathon 2026

- Finished Top 10 nationally; built NETRA, an AI growth copilot for Indian merchants with privacy-preserving market intelligence using k-anonymity and differential privacy.

Published Researcher at ICML (International Conference on Machine Learning) 2026

- Co-authored **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding**, accepted at ICML 2026 (~26% acceptance rate); published as a second-year undergraduate.

Honoured — Award of Excellence 2025

- Honored with the Academic Excellence Award, presented by Shri Suresh Jain (National Organization Secretary, Bharat Vikas Parishad) and Shri Vijay Kumar (Joint Commissioner of Police).

Finalist — ET Gen AI Hackathon 2026

- Co-built an Autonomous Multi-Agent AI Code Intelligence & Remediation Platform; ranked Top Performing out of 3000+ candidates nationwide.